

NPFL087 Statistical MT

Course Structure and Requirements

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unless otherwise stated

Course Outline

1. Metrics of MT Quality.
2. Approaches to MT. SMT, PBMT, NMT, NP-hardness.
3. NMT (Seq2seq, Attention. Transformer).
4. Parallel texts. Sentence and word alignment. hunalign, GIZA++.
5. PBMT: Phrase Extraction, Decoding, MERT. Moses.
6. Morphology in MT. Factors or segmenting, data or linguistics.
7. Syntax in SMT (constituency, dependency, deep).
8. Transformer (and syntax in NMT). LLM-Based Translation.
9. Towards Understanding: Word and Sentence Representations.
10. Multi-Lingual MT, Transfer Learning.
11. Multi-Modal Translation.
12. **Project presentations.**

Related Classes

Informal prerequisites:

- NPFL125 Introduction to Language Technologies
- NPFL070 Language Data Resources

Recommended:

- NPFL114 Deep Learning
- *NPFL140 Large Language Models*

Course Materials (1/2)

Slides:

<https://ufal.mff.cuni.cz/courses/npfl1087>

Videlectures & Wiki of SMT:

<http://mttalks.ufal.ms.mff.cuni.cz/>

Books and others:

- Ondřej Bojar: Čeština a strojový překlad. ÚFAL, 2012.
 - Philipp Koehn: Statistical Machine Translation. Cambridge University Press, 2009. Slides: <http://statmt.org/book/>
- Neural Machine Translation:
<https://arxiv.org/abs/1709.07809>
<http://mt-class.org/jhu/assets/nmt-book.pdf>

Course Materials (2/2)



Kocmi Tom, Macháček Dominik, Bojar Ondřej (2021). **The Reality of Multi-Lingual Machine Translation**. ISBN 978-80-88132-11-0.

<https://ufal.mff.cuni.cz/books/2021-kocmi>

Other good sources:

- <http://mt-class.org/> (UEDIN is updated to NMT.)
- CMU (Graham Neubig) class:
<http://phontron.com/class/mtandseq2seq2017/>
- <http://www.deeplearningbook.org/>
by Goodfellow, Bengio, and Courville.

Grading

Key requirements:

- Work on a project (alone or in a group of two to three).
- Present project results (~30-minute talk).
- Write a report (~4-page scientific paper).

Contributions to the grade:

- 10% participation and homework (possibly CodEx exercises),
- 30% written exam,
- 50% project report,
- 10% project presentation.

Final Grade: $\geq 50\%$ good, $\geq 70\%$ very good, $\geq 90\%$ excellent

Speech Translation: ELITR Demo Videos

- <https://elitr.eu/elitr-complementing-interpreters-and-starting-to-take-notes-for-you>
ELITR final blogpost.
- <http://ufallab.ms.mff.cuni.cz/~bojar/elitr/720p.mp4>
Martin Popel talking Czech; short snippets
- <https://elitr.eu/a-tireless-interpreter/>
Very short demo for Open Doors Day.

Project Suggestions

<https://tinyurl.com/npf1087-2026>